GRADED GROUP PROJECT

General Guidelines

Python for Data Science | APM_51658_EP | M1 AMS

Team size: 4 students **Project period:** W8–W14 **Grade weight:** 20%

1. Purpose

Work in a team of four to investigate a data-science problem using Python. You will study the data or experimental setting, choose and justify suitable methods, conduct experiments, evaluate what worked and what did not, and communicate your conclusions. The emphasis is on the reasoning and quality of the work, rather than simply achieving the best numerical results.

2. Project topics and teamwork

Seven project topics will be available. Each group will work on one topic described in the accompanying *Project Briefs*. The topics may involve different types of data and methods, but the expectations are adapted to each project's objectives.

Every team member must contribute to writing or adapting Python code, understand the complete project, participate in technical decisions and speak during the presentation. The grade is collective: all four members receive the same project grade.

Each team must create a GitHub repository for the project and give both instructors access using the GitHub accounts provided in class. All four students must contribute meaningfully and comparably.

Choosing an advanced topic: Image processing is introduced in W11, and neural networks/CNNs in W12. Projects involving these methods may require additional independent learning early on. Previous experience is helpful, but not a requirement.

3. Session format

W8: The two-hour practical session is entirely dedicated to the project kickoff.

W9–W13: Each practical session begins with about 30–45 minutes of lecture-related exercises; the remaining time is for project work.

W14: Project presentations.

4. Common expectations and methods

Some projects emphasize data preparation, while others focus more on modeling or controlled experiments. You are assessed on how appropriate and well justified your work is for your problem, not on the complexity or number of algorithms used.

All teams must cover the following stages, as relevant to their chosen project:

1. **Define the problem.** Explain what you are trying to predict, discover or investigate, and why the question is relevant.
2. **Understand and explore the data.** Where applicable, describe what each observation represents, examine important variables or classes, and include informative statistics and figures. For other projects, explain the experimental setting.
3. **Prepare the data.** Explain relevant choices about data representation, preprocessing and training/validation/test splits, or how experiments are configured when data preparation is not central.

4. **Investigate suitable methods.** Try a small number of justified approaches or variations; explain what you changed and why.
5. **Evaluate and discuss results.** Choose suitable criteria, compare results fairly, investigate errors or limitations, and acknowledge unsuccessful experiments.
6. **Make the work reproducible.** Organize the code so another student can follow the steps and reproduce the main figures and results.

Unless the *Project Brief* specifies otherwise, no particular model architecture is required. More complicated is not automatically better.

Optional extension: For projects involving a dataset, once the main project is complete, you may identify a second relevant dataset yourself and investigate how well one of your approaches works on it. This can require dataset-specific preprocessing and retraining or fine-tuning. A second dataset is not mandatory, and including one does not automatically lead to a higher grade.

Good methodological choices, fair experiments, reproducibility and clear interpretation matter more than obtaining the best numerical results.

5. Oral presentation

Each group has **15–20 minutes to present** its project, followed by **5 minutes of questions**. Every member must speak. Show the problem, the key choices, the experimental evidence, and the main conclusions.

6. Final submission

Submit one set of deliverables per team. Exact deadlines will be announced.

On GitHub:

- **Code and supporting files.** Include readable Python notebooks (`.ipynb`) and/or scripts (`.py`), with the files needed to run the project (other than large public datasets). Add a short `README.md` with the data source, execution order and instructions.
- **Python packages.** Include a `requirements.txt` listing the libraries used, with versions. You may work in Google Colab or locally; explain briefly in the `README.md` how to run the code.

> **Creating `requirements.txt`.** In a terminal, run `python -m pip freeze > requirements.txt`. In Google Colab, run `!pip freeze > requirements.txt` in a code cell. This may list many preinstalled packages; keep only the libraries your project needs, including their versions. You may also create this text file manually.

On Moodle:

- **Technical report (maximum 4 pages in two-column format or 7 pages in one-column format, excluding references and appendices).** A suggested structure: problem; data or experimental setting; methods; experiments and evaluation; interpretation; limitations and conclusion; references and tool use. Include EDA and data preparation where relevant. You may adapt the structure to your topic.
- **Presentation slides (PDF).** Prepare them for your Week 14 oral presentation, then upload them to Moodle after presenting.

7. Academic integrity and responsible use of tools

- Submit your group's own work. Do not copy existing solutions from Kaggle, GitHub or other sources. Short code examples are allowed if properly cited.
- AI tools may be used for learning and debugging, but not to generate substantial parts of the project.
- Include an "External Resources and AI Usage" section in your report, citing external sources and describing any AI assistance.